Chapter 17

Abdominal Applications of Ultrasound Contrast Agents

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Contrast-Enhanced Sonography

Clinical use of contrast-enhanced sonography (CES) has been shown to reduce or eliminate some current limitations of ultrasound imaging (US) and Doppler blood flow detection.

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Contrast-Enhanced Sonography (Cont.)

- Limitations
  - Spatial and contrast resolution on grayscale
  - Detection of low-velocity blood flow and flow in very small vessels using Doppler flow detection modes

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Vascular Ultrasound Contrast Agents

- Vascular ultrasound contrast agents (UCAs)
  - Enhance Doppler (color and spectral) flow signals by adding more and better acoustic scatterers to the blood stream
  - Improve detection of blood flow from vessels often difficult to assess such as renal arteries, intracranial vessels, small capillaries within organs (i.e., tissue perfusion)

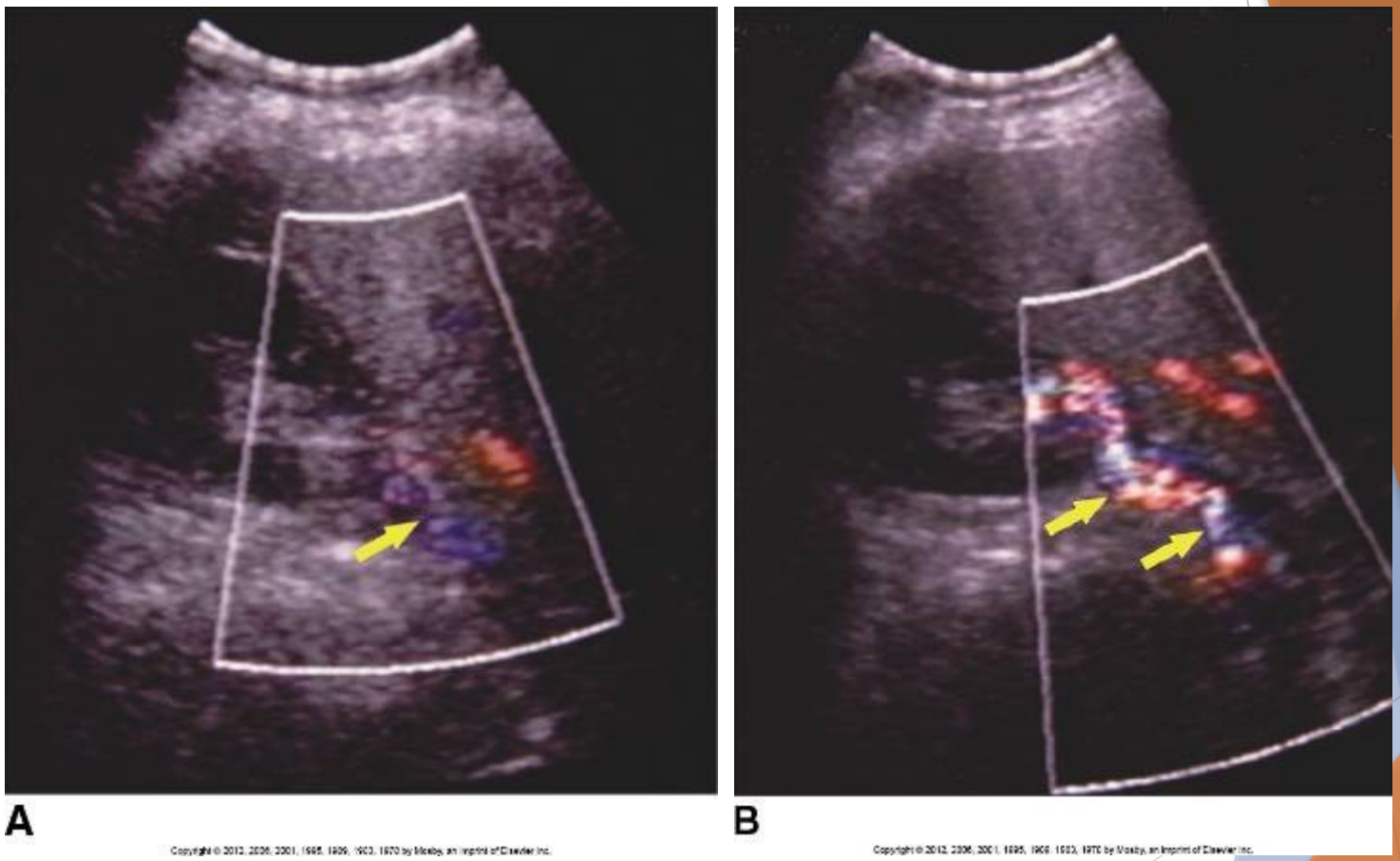
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Vascular Ultrasound Contrast Agents (Cont.)

- Vascular ultrasound contrast agents (UCAs)
  - Improve gray-scale ultrasound visualization of flowing blood and demonstrate changes to grayscale echogenicity of tissues with use of contrast-specific imaging software such as harmonic imaging (HI)

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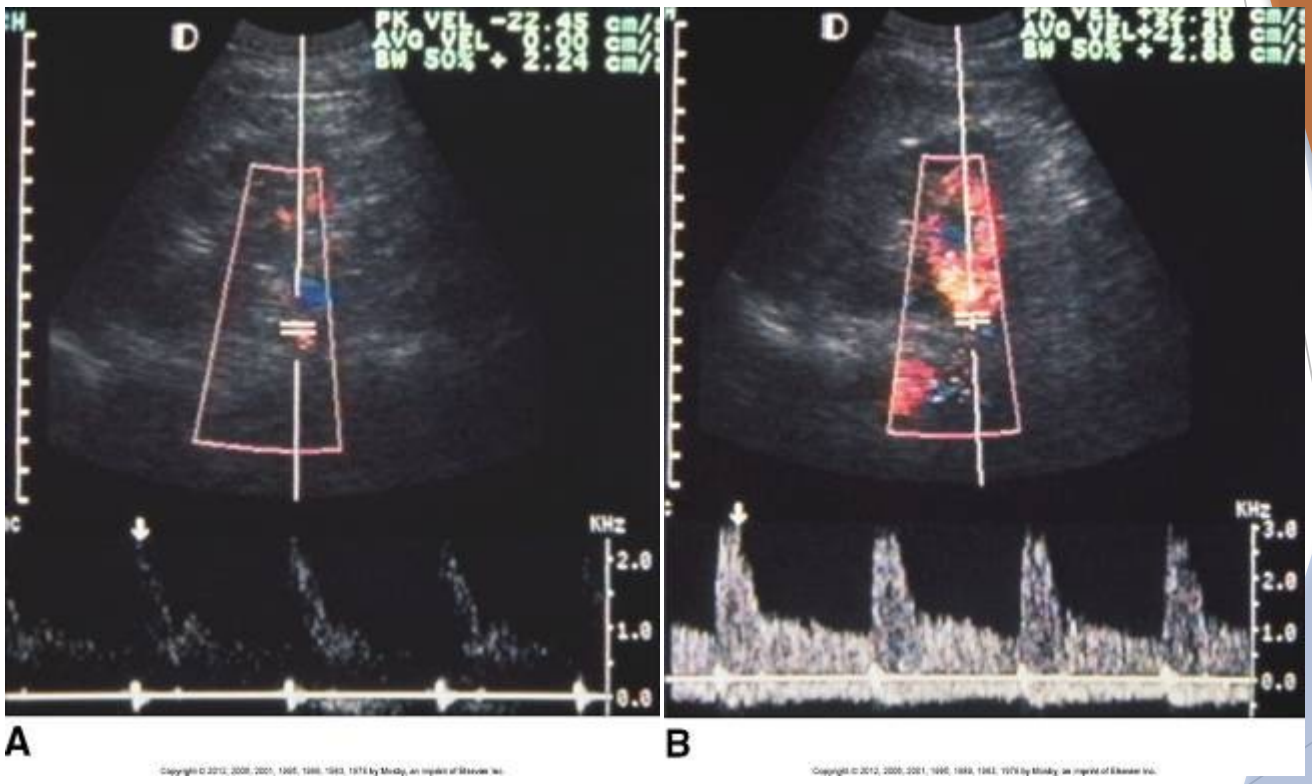
Vascular Ultrasound Contrast Agents (Cont.)



Color Doppler imaging of right renal artery before (**A**) and after (**B**) the administration of a vascular UCA. No vascular abnormality was detected[6]

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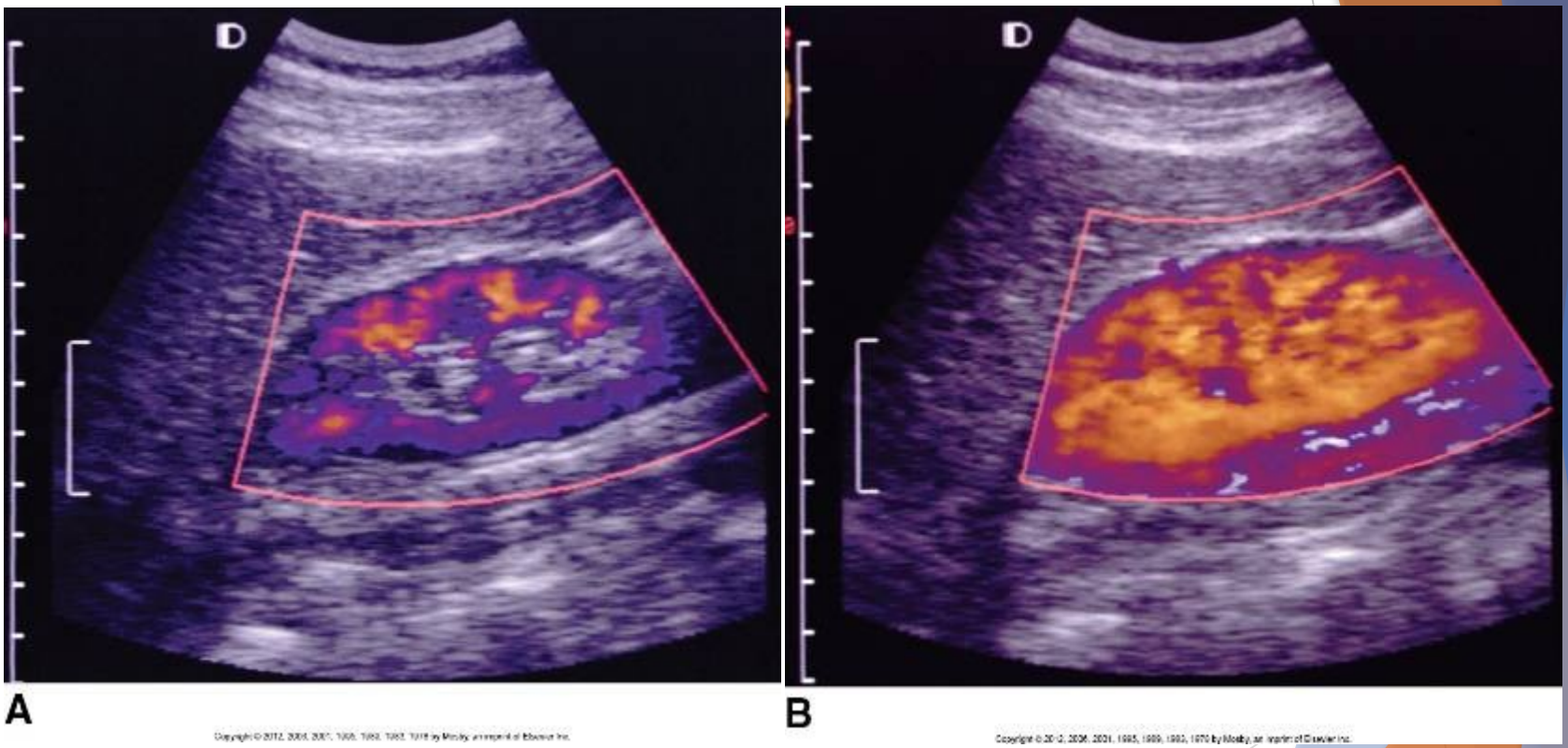
Vascular Ultrasound Contrast Agents (Cont.)



Color-flow imaging and spectral Doppler analysis of renal artery flow. **A,** Before contrast, the spectral waveforms are weak and minimal color-flow information is seen. **B,** After intravenous (IV) administration of a contrast.

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Vascular Ultrasound Contrast Agents (Cont.)



Power Doppler imaging of a normal right kidney before (**A**) and after (**B**) injection of a vascular UCA.

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Vascular Ultrasound Contrast Agents (Cont.)

- In 1968 researchers injected agitated saline directly into ascending aorta and cardiac chambers during echocardiographic examinations.
- Microbubbles formed by agitation resulted in strong reflections arising from within normally echo-free lumen of aorta and chambers of the heart.
- Microbubbles produced by simple agitation are nonuniform in size and relatively large and unstable.
- Unsuitable for sonographic evaluation of the left heart and systemic circulation because microbubbles do not persist through passage of pulmonary and cardiac circulations

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Vascular Ultrasound Contrast Agents (Cont.)

- Clinically useful UCAs must
  - Be nontoxic
  - Have microbubbles or microparticles small enough to traverse pulmonary capillary beds (i.e., less than 8 microns)
  - Be stable enough to provide multiple recirculations

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Vascular Ultrasound Contrast Agents (Cont.)

- FDA-approved for use in echocardiographic applications
  - Definity
  - Optison
  - Imagent
- Other agents in use
  - Levovist (Europe only)
  - SonoVue (Europe only)

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Types of Ultrasound Contrast Agents

- Tissue-specific UCAs
  - Microbubbles of agents removed from the blood pool and taken up by (or have affinity toward) specific tissues such as reticuloendothelial system (RES) in the liver and spleen or thrombus

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Tissue-Specific UCAs

- Over time, the presence of contrast microbubbles within or attached to tissue changes its sonographic appearance.
- By changing the signal impedance of normal and abnormal tissues, agents improve detectability of abnormalities.
- Tissue-specific UCAs are typically administered by intravenous (IV) injection.

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Tissue-Specific UCAs (Cont.)

- Sonazoid
  - Tissue-specific UCA currently approved for use in Japan
  - After being injected intravenously, Sonazoid behaves as a vascular agent (i.e., enhances the detection of flowing blood).
  - Over time, microbubbles are phagocytosed by RES (macrophage Kupffer cells) of the liver and spleen.
  - Intact microbubbles may remain stationary in tissue for several hours.

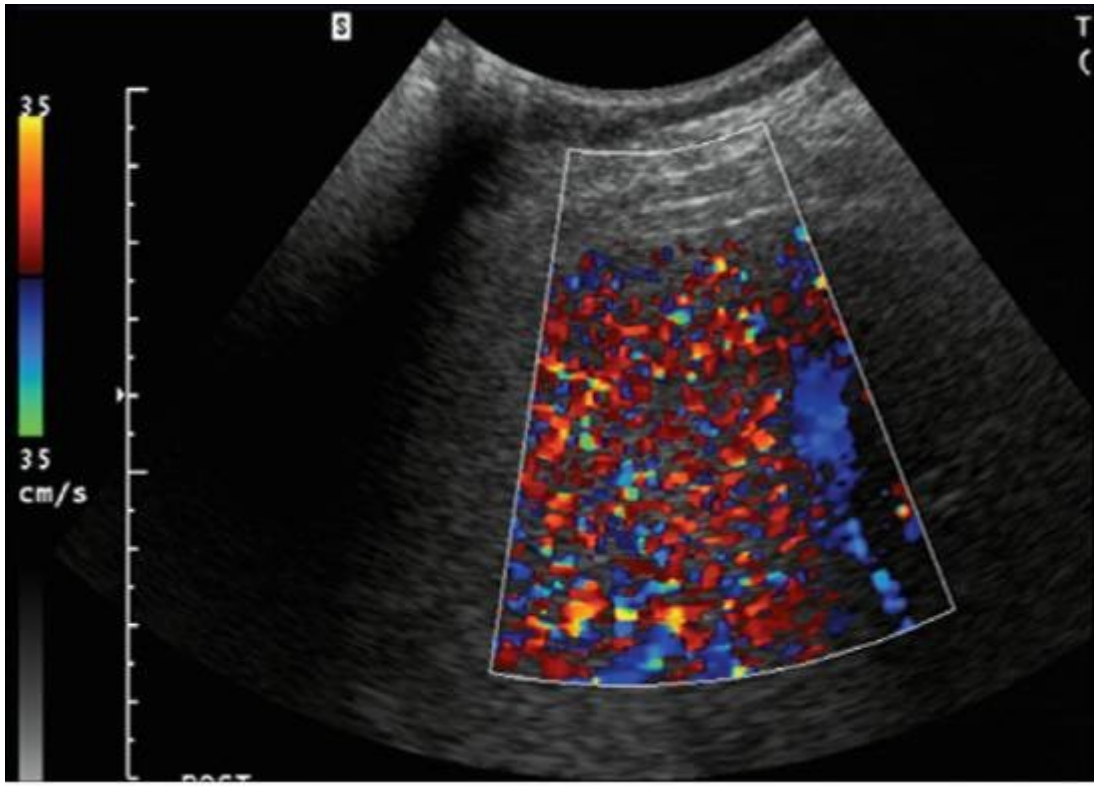
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Tissue-Specific UCAs (Cont.)

- Sonazoid
  - When insonated after uptake, stationary contrast microbubbles increase reflectivity of contrast-containing tissue.
  - If appropriate level of acoustic energy is applied to tissue, microbubbles first oscillate (emitting harmonic signals that can be detected with grayscale HI) and then rupture.
  - Rupture of microbubbles results in random Doppler shifts appearing as transient mosaic of colors on color Doppler display.

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Tissue-Specific UCAs: Sonazoid (Japan)



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Rupture of contrast microbubbles within the RES cells of the liver results in the characteristic random color display.

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Types of Ultrasound Contrast Agents

- Oral UCAs
  - SonoRx™ contains simethicone-coated cellulose as the active ingredient.
  - Ingestion of SonoRx™ results in homogeneous transmission of sound through contrast-filled stomach.

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Ultrasound Equipment Modifications

- Use of “contrast-specific” ultrasound (US) imaging modes has been found to greatly improve clinical utility of CES.
- Because the current US system platforms is digital and software-driven, necessary changes can be implemented relatively quickly.

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Ultrasound Equipment Modifications (Cont.)

- Harmonic imaging (HI) uses same broadband transducers as conventional US.
- In HI mode, US system configured to receive only echoes at second harmonic frequency (twice the transmit frequency).
- When using microbubble-based UCAs, microbubbles oscillate (i.e., get larger and smaller) when subjected to acoustic energy present in US field.

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Ultrasound Equipment Modifications (Cont.)

- Reflected echoes from oscillating microbubbles contain energy components at fundamental frequency, as well as higher and lower harmonics (subharmonics).
- Majority of development efforts involving HI have focused on grayscale HI (GSHI), which allows the detection of contrast-enhancement of blood flow and organs with grayscale US.

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Ultrasound Equipment Modifications (Cont.)

- Use of GSHI for CES avoids many of the limitations and artifacts encountered when using Doppler US techniques and UCAs, such as angle dependence and “color blooming.”

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Ultrasound Equipment Modifications (Cont.)

- In the HI mode, echoes from oscillating microbubbles have a higher signal-to-noise ratio than conventional US.
- Means that regions with microbubbles (e.g., blood vessels and organ parenchyma) are more easily appreciated visually.
- Although body tissues also generate harmonic signals, their contribution to an image is negligible, compared with those from microbubbles.

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Ultrasound Equipment Modifications (Cont.)

- HI technology: image processing algorithms to subtract echoes arising from body tissues; echoes arising from contrast microbubbles preferentially displayed
- Wide-band GSHI provides a way to better differentiate areas with and without contrast and has a potential to demonstrate real-time grayscale blood pool imaging (i.e., “perfusion imaging”).

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Ultrasound Equipment Modifications (Cont.)

- In most cases, benefits of using GSHI for CES (e.g., higher frame rates, improved contrast resolution, reduced artifacts) so great that conventional (non-HI) color-flow imaging is neither necessary nor advised

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Ultrasound Equipment Modifications (Cont.)

- Harmonic Doppler US modes
  - Harmonic power Doppler imaging
  - Power modulation imaging
- Preliminary investigations with harmonic Doppler modes have identified some potential benefits:
  - Enhanced detection of very low velocity blood flow in organs
  - Suppression of color “flash” artifacts

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Ultrasound Equipment Modifications (Cont.)

- When using microbubble-based UCAs, energy present within acoustic field can have detrimental effect on microbubbles.
- Significant number of microbubbles can be destroyed by acoustic pressure even though actual pressure contained within US field is relatively low.

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Ultrasound Equipment Modifications (Cont.)

- Once microbubble destroyed, contrast enhancement no longer provided
- To combat this, use low acoustic output power as defined by mechanical index (MI).
- Reducing MI, however, limits tissue penetration, so not always adequate solution

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Ultrasound Equipment Modifications (Cont.)

- Intermittent imaging capabilities on US systems provide additional option to user seeking to reduce microbubble destruction during contrast-enhanced examinations.
- In this mode, system is gated to only transmit and receive data at predetermined intervals.

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Ultrasound Equipment Modifications (Cont.)

- Intermittent imaging reduces exposure of contrast microbubbles to acoustic energy and allows additional microbubbles to enter field between signal transmissions.

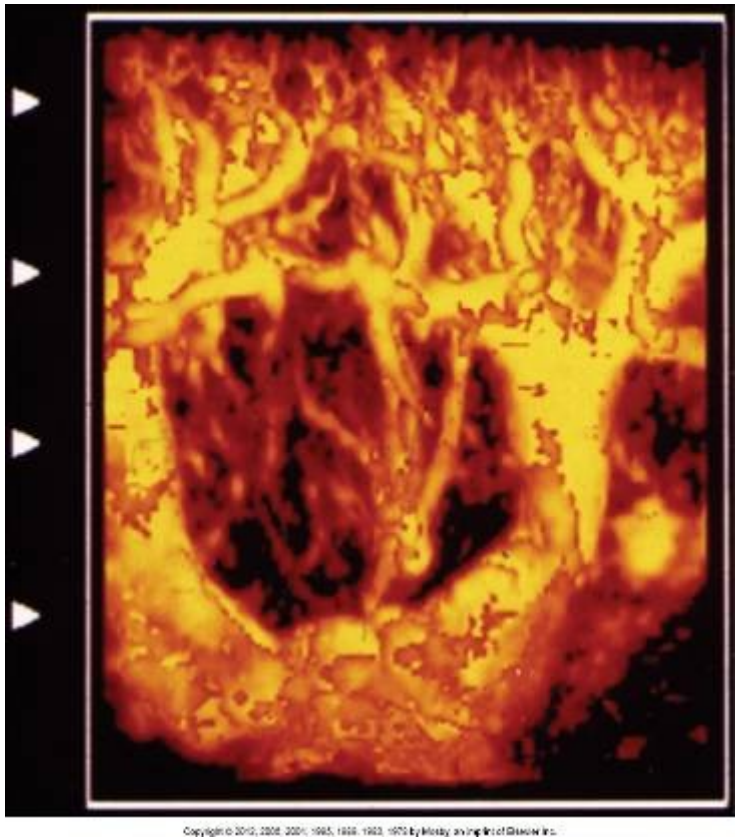
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Ultrasound Equipment Modifications (Cont.)

- Other system modifications in development
  - On-board video densitometry
  - Calculation of integrated backscatter from contrast
  - Advanced three-dimensional (3D) imaging

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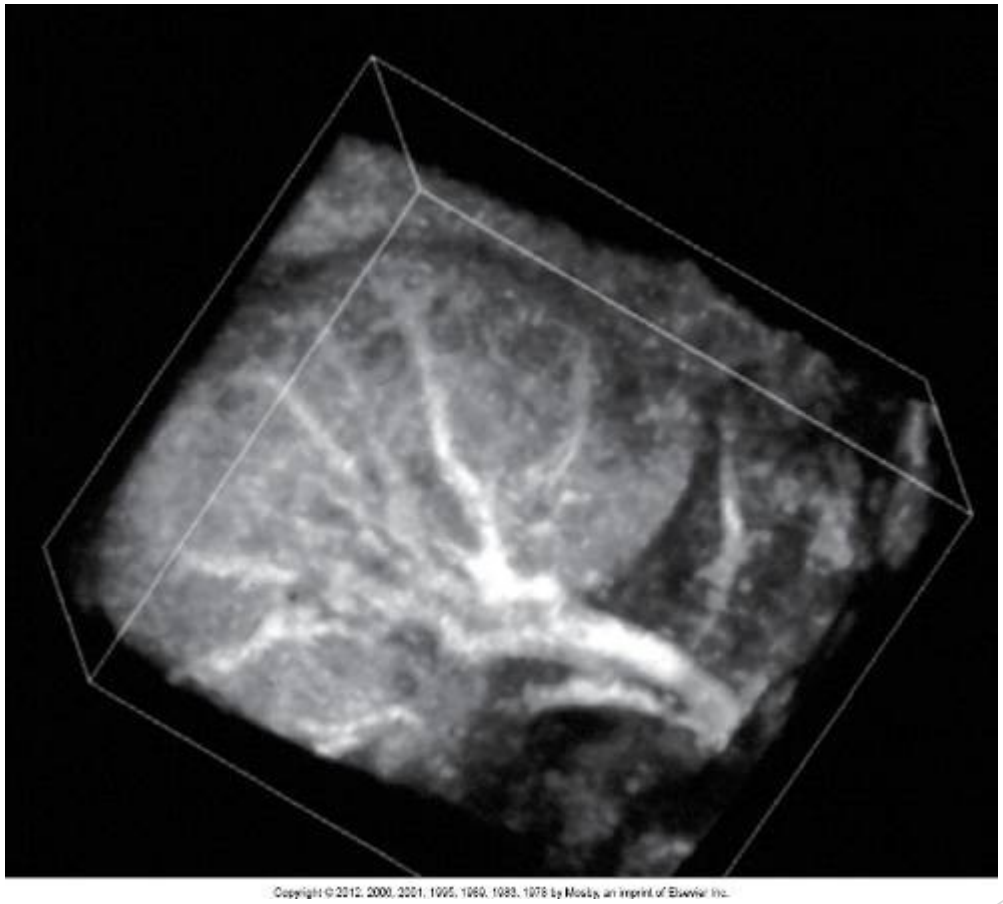
Ultrasound Equipment Modifications (Cont.)



Contrast-enhanced 3D power Doppler image of a kidney in an animal model. Fine detail of the renal vasculature is noted.

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Ultrasound Equipment Modifications (Cont.)


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"Ultrasound angiogram." Combination of a 3D grayscale harmonic image and the administration of a vascular UCA.

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Clinical Applications: Liver

- Sonography usually sensitive for detection of medium-to-large hepatic lesions
- Limited in ability to detect small (<10 mm), isoechoic, and/or peripherally located lesions, particularly in obese patients or patients with diffuse liver

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Clinical Applications: Liver (Cont.)

- Hepatic US blood flow studies limited by low velocity blood flow and in detection of flow in intrahepatic artery branches
- US contrast agents improve accuracy of hepatic sonography:
  - Enhanced detection and characterization of hepatic masses
  - Improved detection of intra- and extrahepatic blood flow

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Clinical Applications: Liver (Cont.)

- Vascular UCAs improve detection of hepatic blood flow in both normal subjects and patients with liver disease and portal hypertension.
- Assessment of flow through transjugular intrahepatic portosystemic shunts (TIPS).

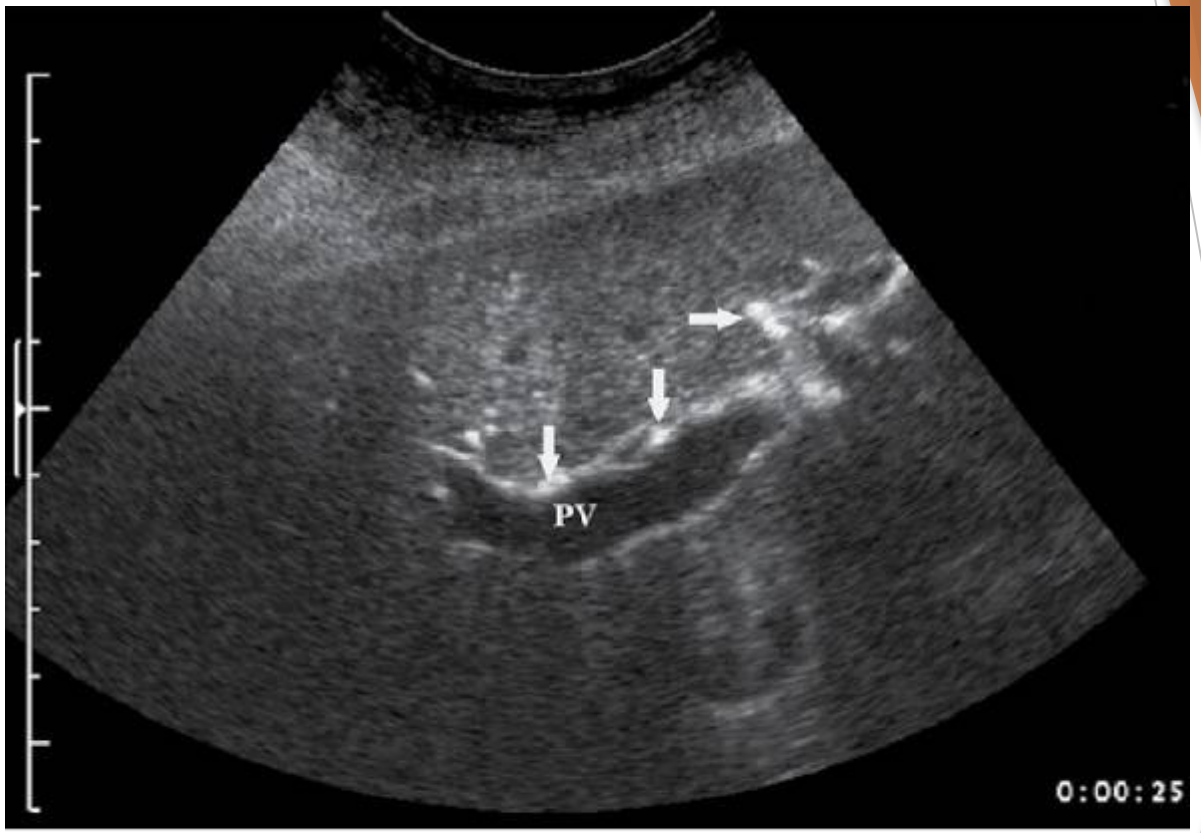
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Clinical Applications: Liver (Cont.)

- Using CES possible to distinguish various phases of blood flow to and within liver
- In normal situations after IV administration of UCA contrast-enhanced flow in hepatic artery identified first (arterial or early vascular phase), followed by enhanced portal venous flow (portal venous phase)
- Detection of flow in hepatic capillaries identified later (late vascular phase) as parenchymal blush

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Clinical Applications: Liver (Cont.)



A

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Twenty-five seconds after IV injection of an UCA, first vessels to demonstrate contrast enhancement in the liver are the hepatic arteries (*arrows*). Portal vein remains anechoic in the arterial phase.

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Clinical Applications: Liver (Cont.)



B

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Portal venous phase occurred 37 seconds after injection with contrast seen in the portal vein (PV).

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Clinical Applications: Liver (Cont.)



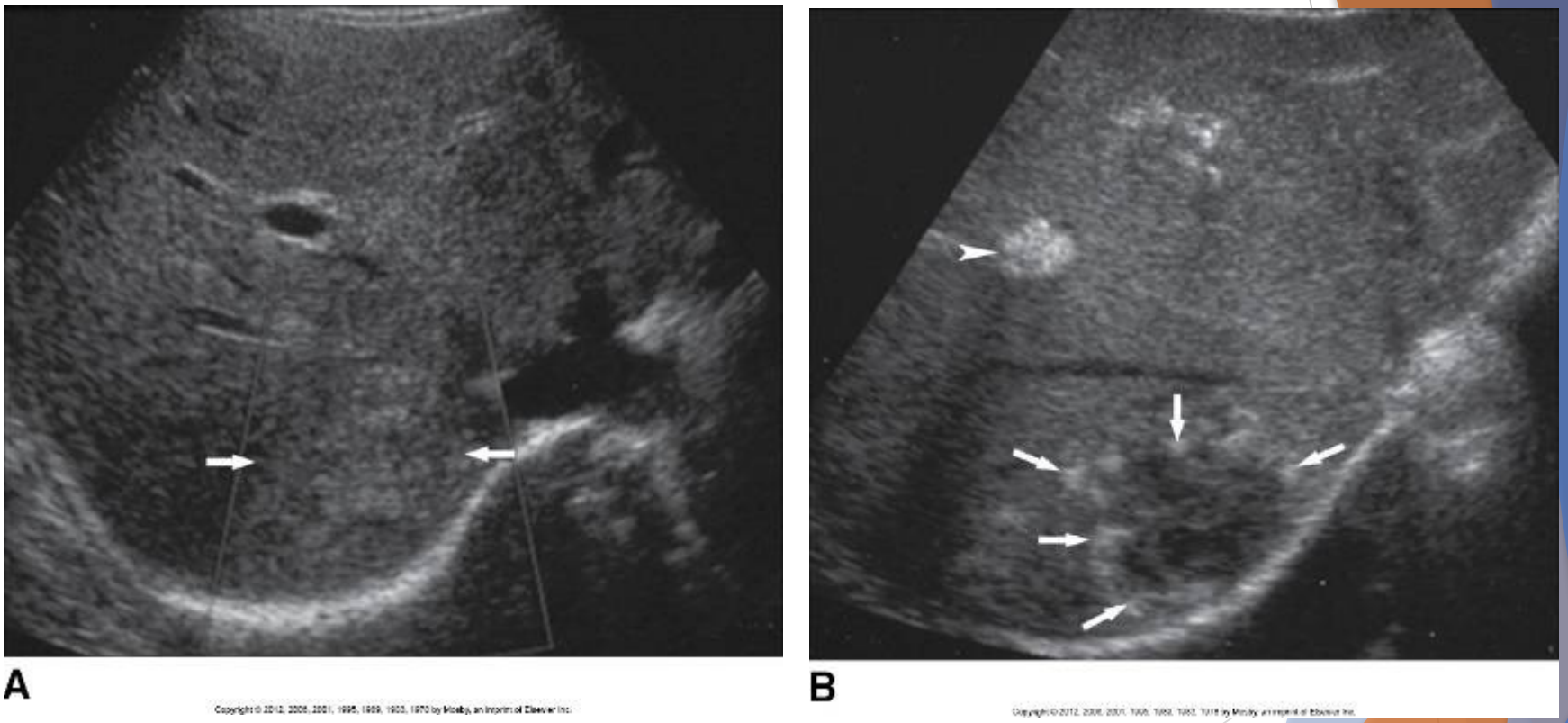
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Late vascular phase at 71 seconds after injection, liver parenchyma is becoming echogenic and enhancement of the major hepatic vessels is seen (*arrowheads*).

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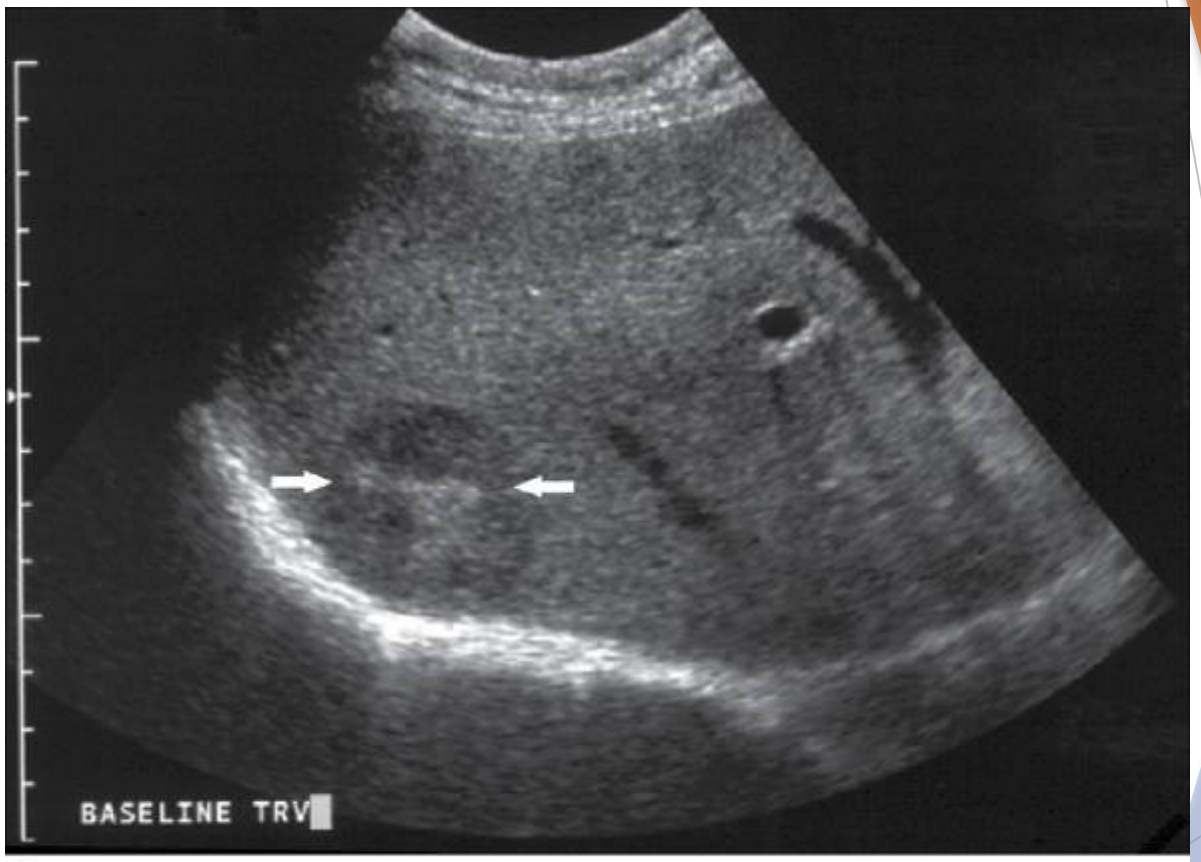
Clinical Applications: Liver (Cont.)



Hemangioma.

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Clinical Applications: Liver (Cont.)



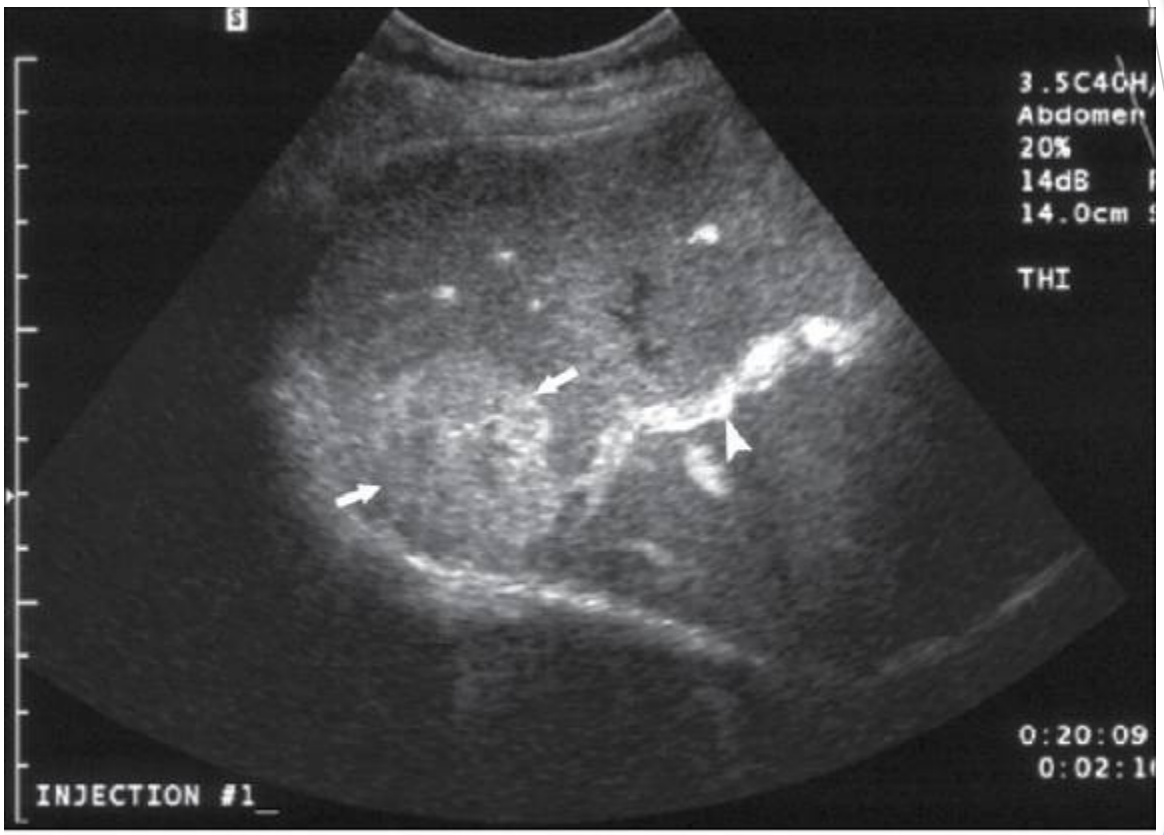
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Hepatocellular carcinoma on conventional ultrasound.

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Clinical Applications: Liver (Cont.)



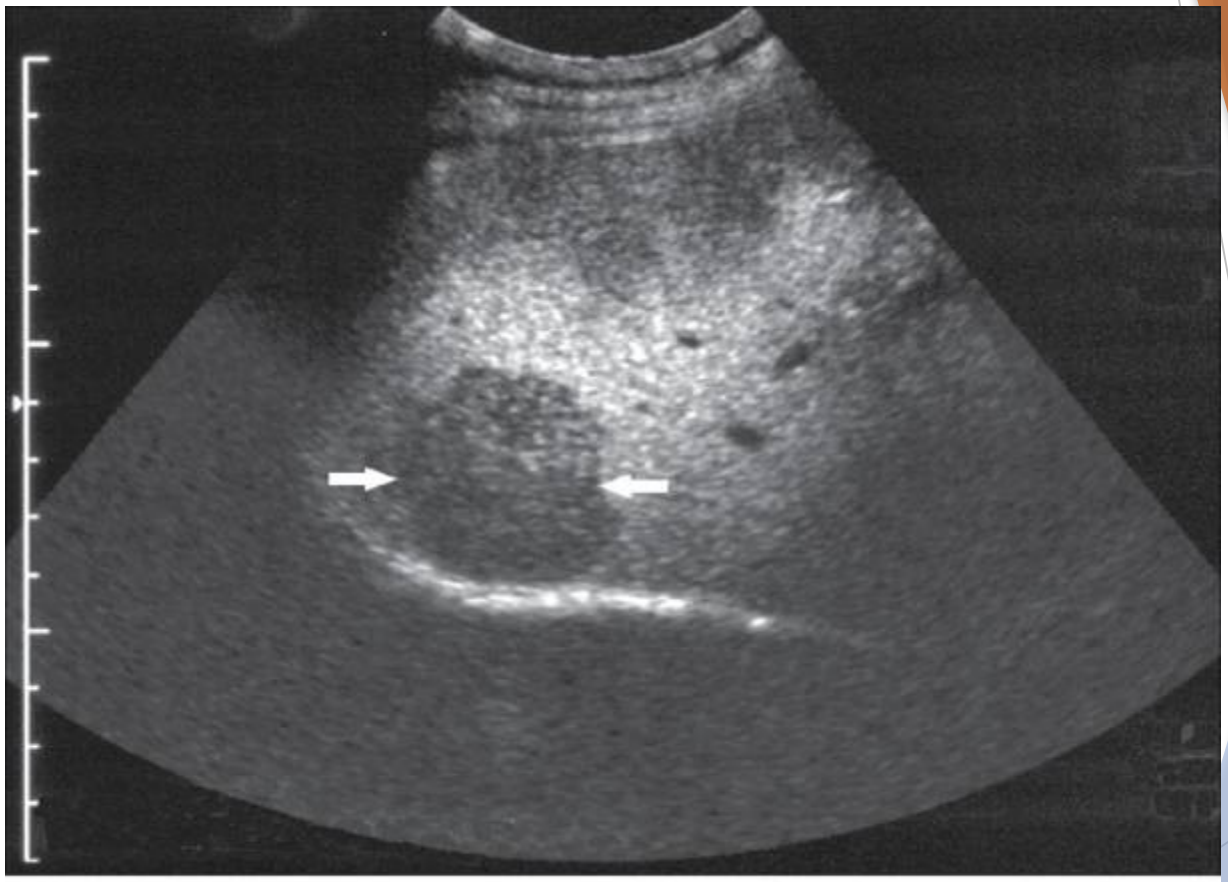
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HCC after contrast administration using GSHI.

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Clinical Applications: Liver (Cont.)



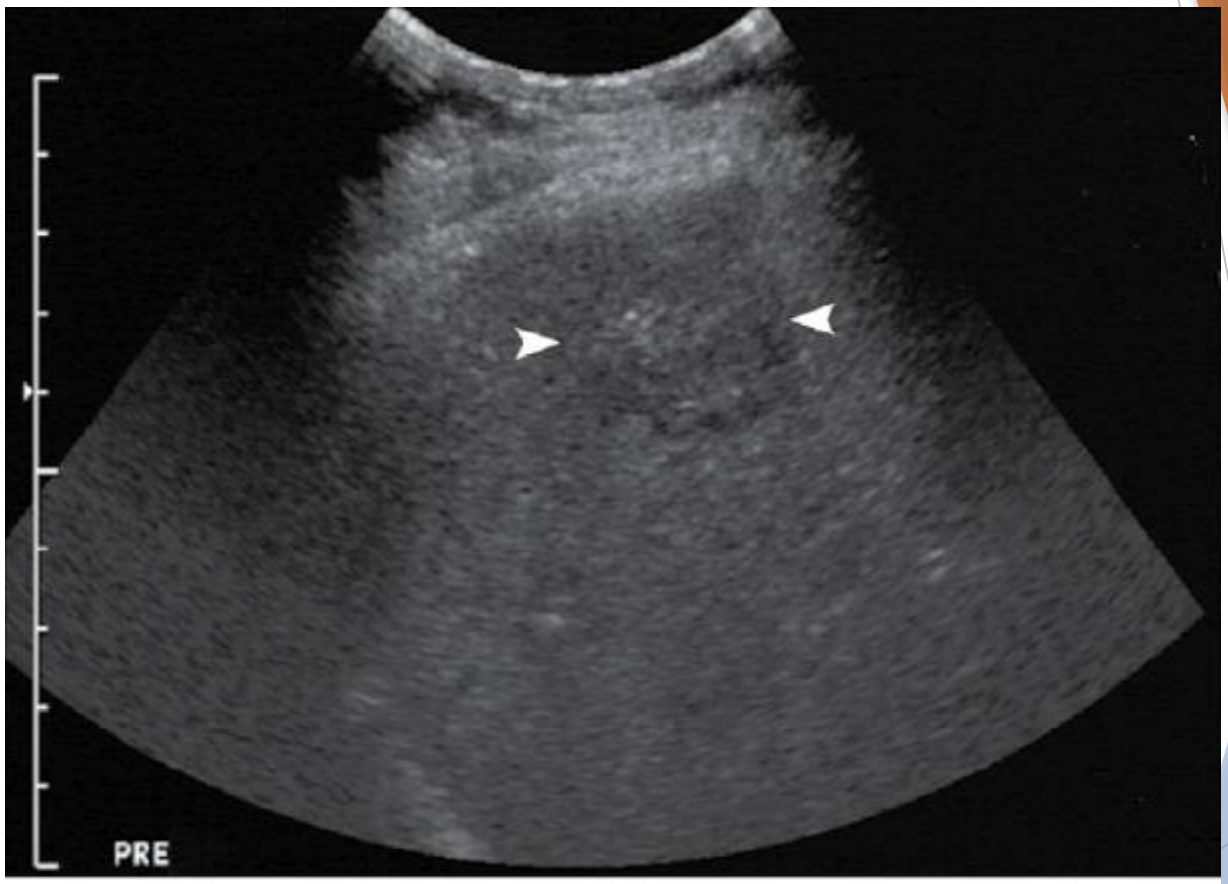
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HCC. Lesion on delayed imaging.

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Clinical Applications: Liver (Cont.)



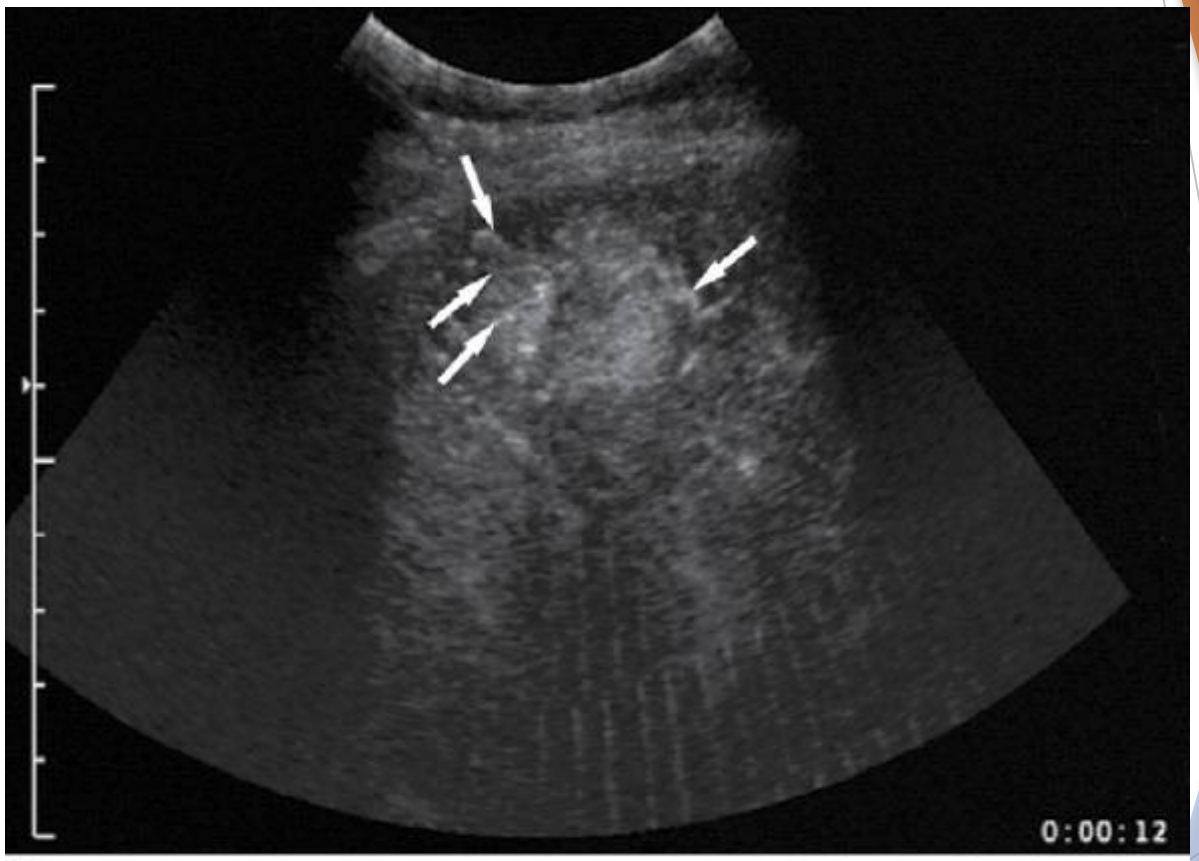
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Focal nodular hyperplasia on conventional ultrasound.

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Clinical Applications: Liver (Cont.)

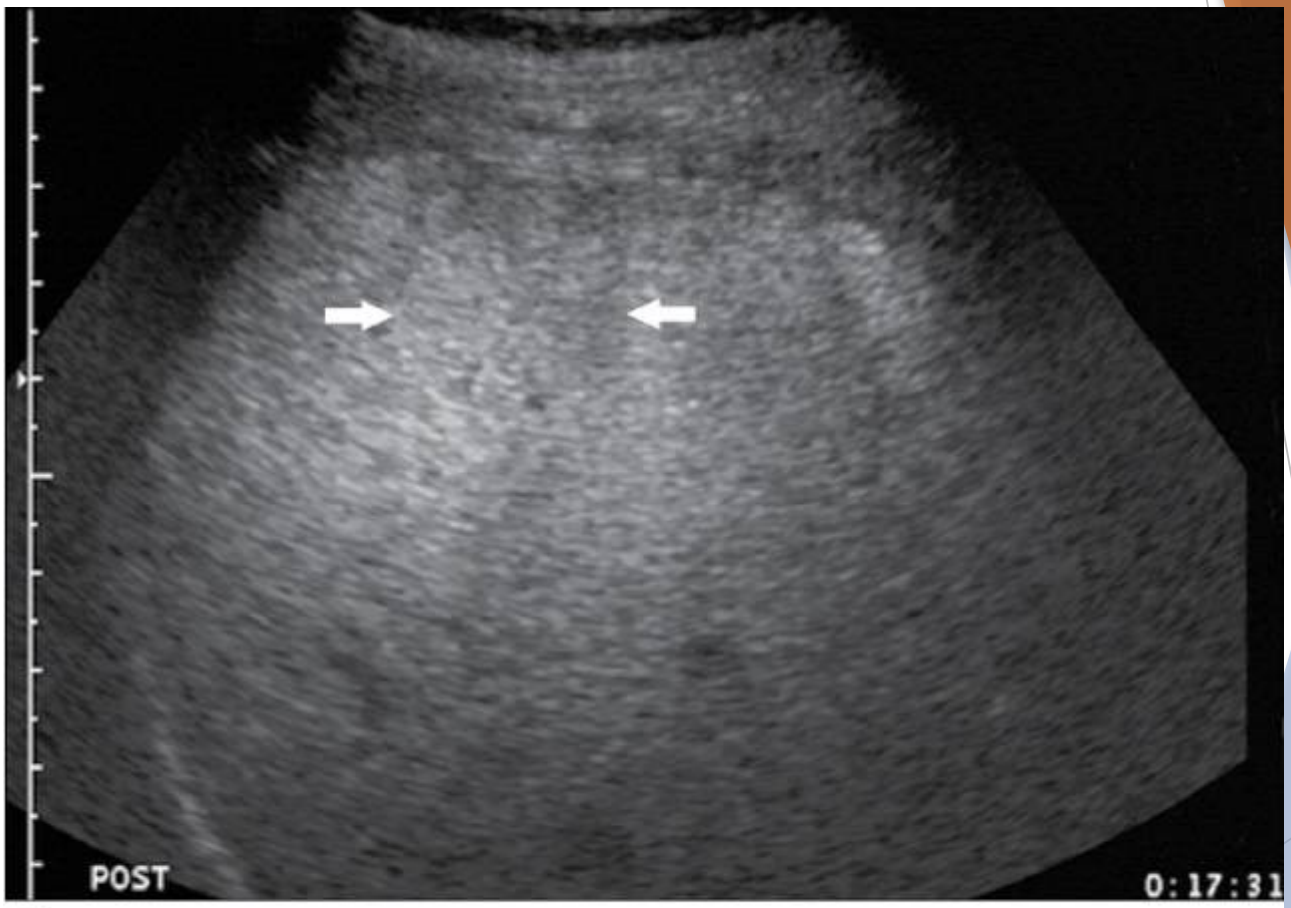


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Focal nodular hyperplasia after contrast administration using GSHI.

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Clinical Applications: Liver (Cont.)

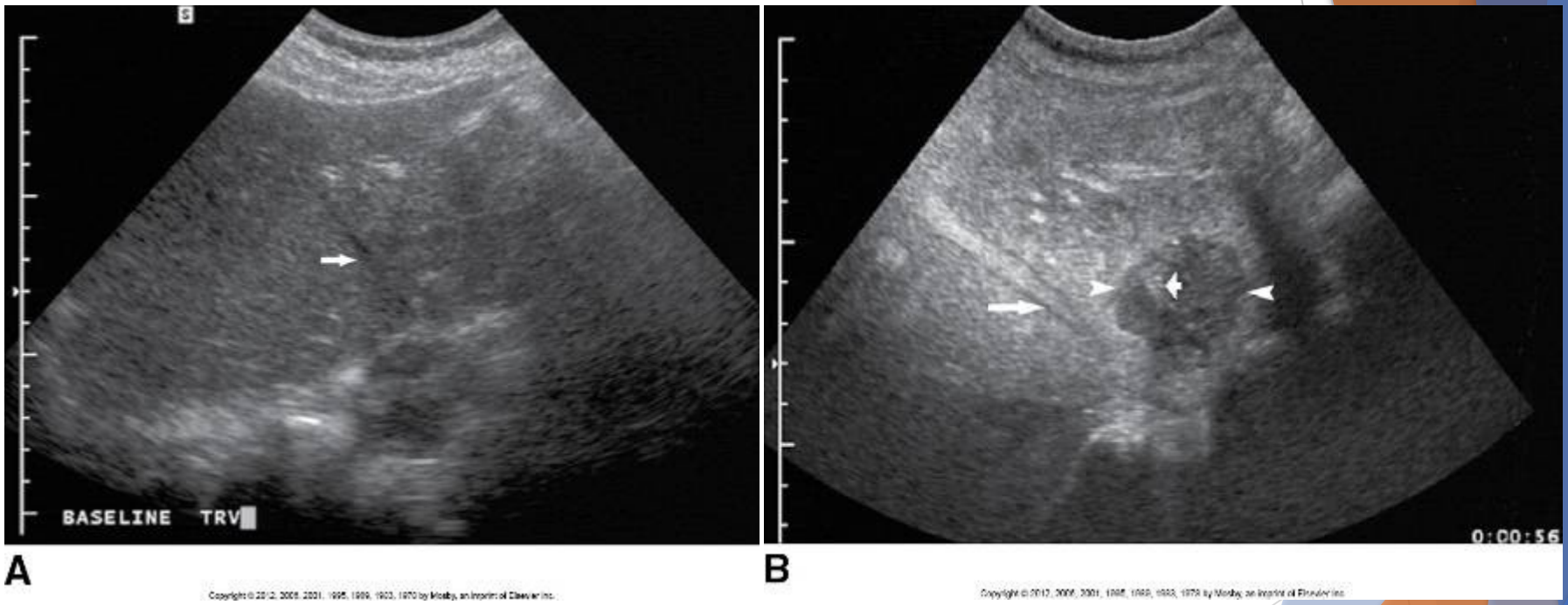


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Focal nodular hyperplasia on delayed imaging.

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Clinical Applications: Liver (Cont.)



Improved detection and delineation of a colorectal carcinoma liver metastasis.

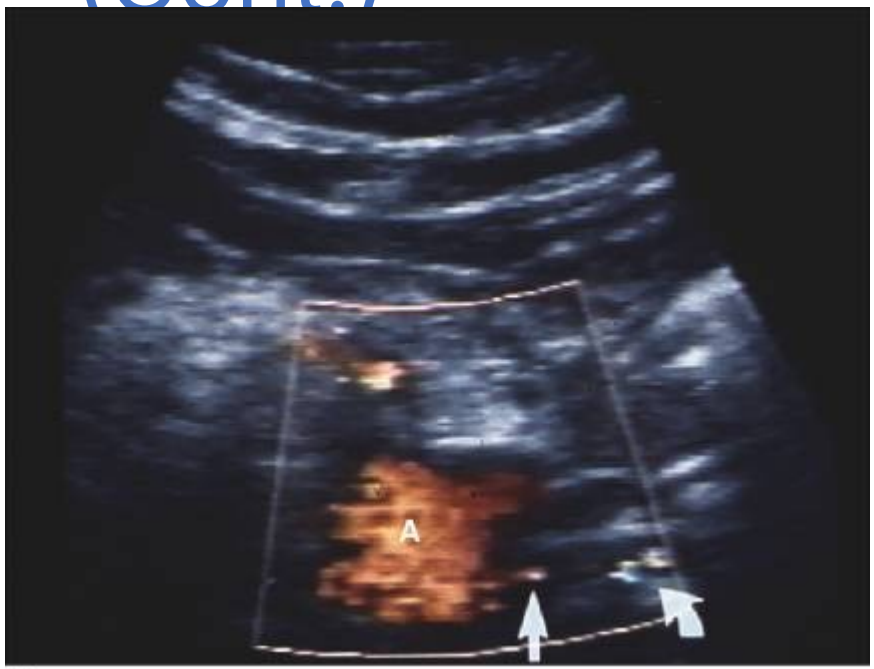
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Clinical Applications: Renal

- Vascular contrast agents significantly increase ability to visualize blood flow using color-flow imaging and improve intensity of spectral Doppler flow signals.
- When renal arteries are not visualized or spectral waveforms are of poor quality, IV administration of contrast can improve the examination process and potentially reduce the number of technically inadequate or otherwise nondiagnostic examinations.

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Clinical Applications: Renal (Cont.)



A



B

A, Before intravenous administration of contrast flow in the aorta (A) and proximal-most renal artery (*arrow*) visualized, there is aliasing of the color-flow display (*curved arrow*). **B,** After injection of contrast, the stenotic vessel lumen (*arrows*) can be clearly visualized.

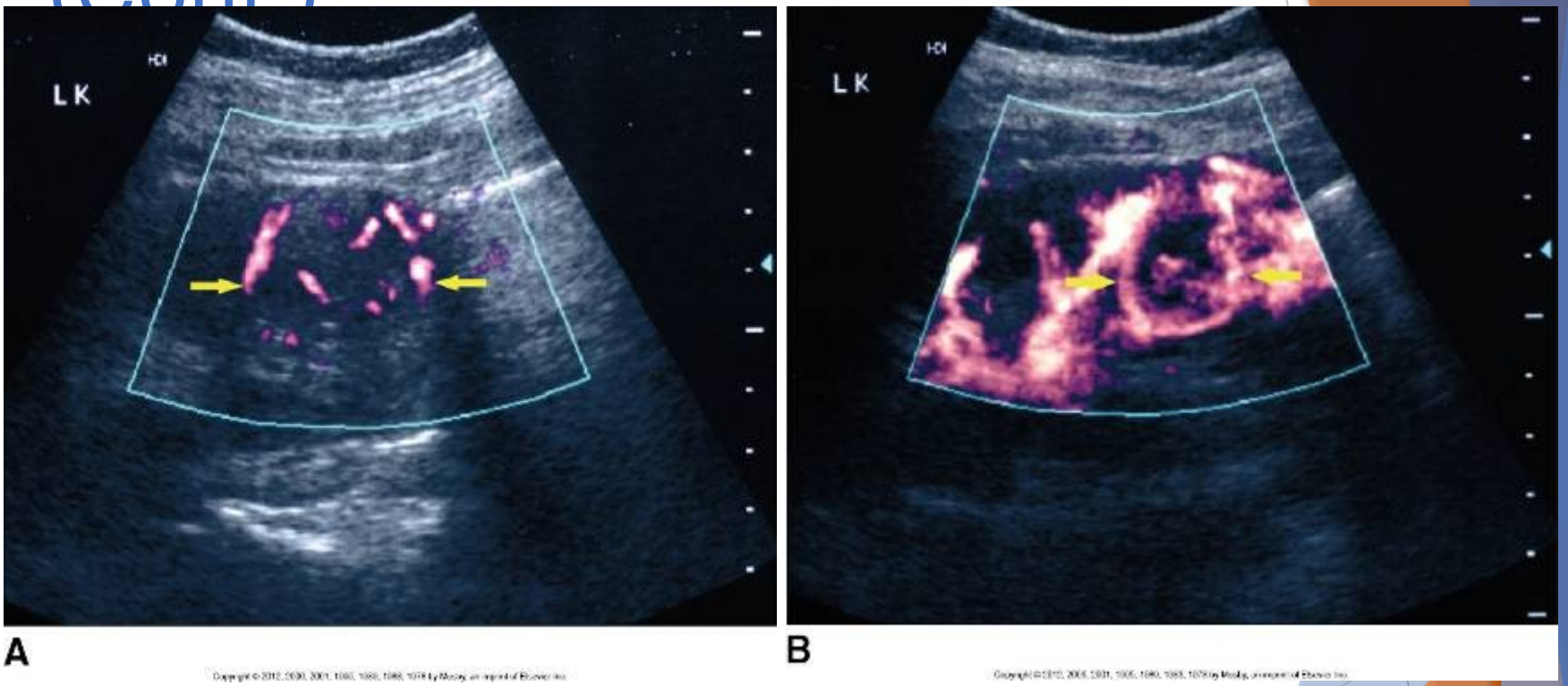
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Clinical Applications: Renal (Cont.)

- Benefits of UCAs in renal sonography
  - Identify and demonstrate normal renal vasculature
  - Detect presence of lesions that distort vascular architecture
  - Improve detection of abnormal vessels present in renal tumors
  - Identify flow voids that result from tumor thrombus in cases of renal vein or IVC involvement

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Clinical Applications: Renal (Cont.)



A, Precontrast. Subtle hypoechoic mass identified near lower pole of the left kidney. **B,** After injection of contrast, significantly more flow signals are detected, including intratumoral flow. Confirmed renal cell carcinoma.

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Clinical Applications: Spleen

- Clinical investigations underway for the use of CES for
  - Hodgkin lymphoma
  - Malignant nodules
  - Splenic trauma
  - Other splenic tumors

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Clinical Applications: Pancreas

- Clinical investigation underway for use of CES for
  - Pancreatic tumors
  - Pancreatitis
  - Parapancreatic lesions

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Clinical Applications: Organ Transplants

- Enhanced detection of blood flow provided by CES improves assessment of blood flow in arteries and veins that supply transplanted organ and vessels to which vessels anastomosed.
- CES also found to improve ability to detect lack of flow within transplanted organs (i.e., ischemic regions) within renal and pancreatic grafts

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Clinical Applications: Organ Transplants (Cont.)

- Contrast-enhanced, contrast flow imaging (CFI) significantly improves detection of hepatic arterial flow in transplant recipients.
- Contrast-enhanced sonography also superior to conventional sonography for differentiating parenchymal abnormalities (e.g., acute tubular necrosis [ATN]) from other abnormalities.

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Clinical Applications: Other Applications

- CES may also be used to evaluate
  - Aortic stent-graft
  - Mesenteric arteries for ischemia
  - Aortic aneurysms, stenoses, endovascular leaks or dissections
  - Inferior vena cava (IVC) for filter or thromboses
  - Tumor size before and after ablation

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